

Univariate Data Analysis Report

Dataset: Student Placement Performance Profiles

1. Executive Summary

This report provides a preliminary **Univariate Data Analysis** focused on the quantitative metrics of 215 student profiles. The analytical focus spans academic performance benchmarks across different schooling stages (10th grade, 12th grade, undergraduate level), entrance test results, MBA grade distributions, and professional corporate compensation packages.

2. Central Tendency Matrix

Metric	ssc_p (10th %)	hsc_p (12th %)	degree_p (%)	etest_p (%)	mba_p (%)	salary (INR)
Mean	67.30%	66.33%	66.37%	72.10%	62.28%	■288,655.41
Median	67.00%	65.00%	66.00%	71.00%	62.00%	■265,000
Mode	62.00%	63.00%	65.00%	60.00%	56.70%	■300,000

3. Core Interpretations & Insights

- **Data Integrity Alert (Mode Distortions):** An audit of the base Univariate script reveals structural calculation bugs affecting the 'Mode' row. Because pandas .mode() functions return dataframes to accommodate multimodal distributions, unhandled horizontal row binding has forced arbitrary index alignments (e.g., pulling structural indices like '1' into the metric space). A custom iterative parser has been introduced to resolve this issue.
- **Academic Score Symmetry:** Academic benchmarks across 10th grade (ssc_p Mean: 67.30%), 12th grade (hsc_p Mean: 66.33%), and undergrad degrees (degree_p Mean: 66.37%) display stable median-to-mean convergence. This points to a standard normal bell curve profile with highly predictable student grading tracks.
- **The Higher Education Compression:** Students experience tougher performance curves during postgraduate tracks, with the mba_p average dipping to 62.27%. This indicates strict post-graduate assessment parameters.
- **Salary Dispersion & Outliers:** Corporate compensation exhibits classic right-skewed behavior. The overall average salary (Mean: ■288,655) is significantly inflated above the midpoint baseline (Median: ■265,000). This gap indicates that a small, highly compensated cluster of elite outlier selections is pulling up the broader institutional placement average.